Department of Computer Science and Engineering

Rajshahi University

Part-IV Odd Semester Examination-2025

CSE-4192: Blockchain Lab.

Answer all the Questions.

Marks: 17.50

1. Write a program in Python to prove membership and non-membership in a Merkle Tree Blockchain. 7.5

2. *Answer the following questions:* 10

   a) Run the blockchain network using:

      Chaincode name: **mychaincode**

      Channel name: **mychannel**

   b) Create the following asset records on the ledger:

ID	Asset Name	Price (BDT)	Owner
ast1	Dell Laptop	65000	Alice
ast2	Toyota Car	1850000	Alice
ast3	Leather Backpack	3200	Alice
ast4	Mountain Bicycle	28000	Alice
ast5	Apartment (2BHK)	6500000	Alice

Hence:

i) Query and display all assets stored in the blockchain ledger.

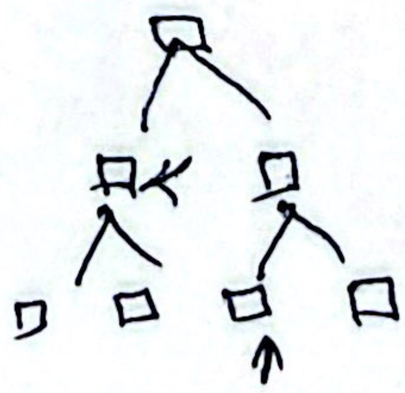
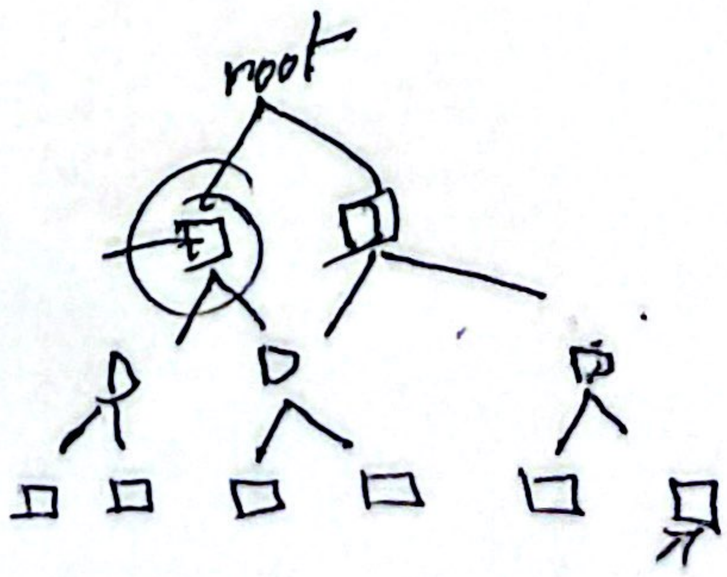
ii) Transfer the ownership of the Dell Laptop (ast1) from Alice to Bob.

iii) Transfer the ownership of the Toyota Car (ast2) from Alice to Charlie.

iv) Update the price of the Dell Laptop (ast1) to 80000 BDT.

v) Query and display the transaction history of asset ast1.

vi) Display the channel name of the running blockchain network.

vii) Attempt to transfer ownership of a non-existing asset (e.g., ast10) and observe the system response.

Explain the error message.